Ecological statistics on viral abundance and diversity

March 25, 2026

Run by Sabrina Sadiq

Download the R scripts and data sets here!

github.com/sabrinasadiq/velab-workshop-resources/

TODAY'S WORKSHOP

What are you aiming to get out of today?

- Want to learn a new concept?
- Baseline script for statistics in R?
- Inspiration for your next project?
- Advice on a current project?

TODAY'S WORKSHOP AGENDA

I didn't have time to write a short workshop, so I wrote a long one instead

Concepts and definitions

Example data sets

Setting up your metadata

- Final list of viral contigs
- Blast results (taxonomy) + RSEM (readcounts)
- Library properties

Quick glimpse of your data set

- Virome composition (1 figure)
- Abundance & diversity (1 figure)

Detailed ecological statistics

- Ecological properties vs abundance and alpha diversity
- Best fit model finding
- Beta diversity analysis

ECOLOGICAL ANALYSES OF RNA VIRAL DATA SETS

Concepts and definitions

1. Abundance
2. Alpha diversity
3. Beta diversity

Viral read abundance

$$\frac{\text{Number of reads aligning to RNA viruses}}{\text{Total number of reads in the library}} \times 100$$

Relative viral abundance

$$\frac{\text{Number of reads aligning to a viral taxa}}{\text{Total number of } \textbf{viral} \text{ reads in the library}} \times 100$$

ECOLOGICAL ANALYSES OF RNA VIRAL DATA SETS

Concepts and definitions

1. Abundance
2. Alpha diversity
3. Beta diversity

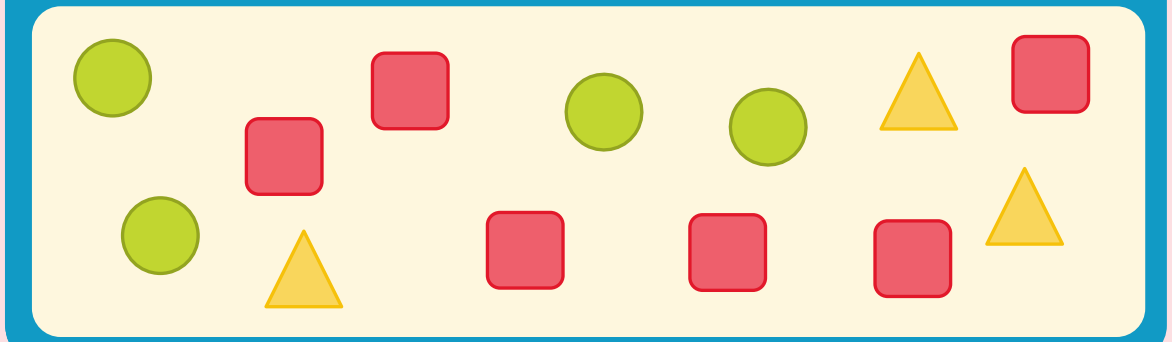
Viral read abundance

$$\frac{\text{Number of reads aligning to RNA viruses}}{\text{Total number of reads in the library}} \times 100$$

Relative viral abundance

$$\frac{\text{Number of reads aligning to a viral taxa}}{\text{Total number of } \textbf{viral} \text{ reads in the library}} \times 100$$

Alpha diversity | diversity *within* a community



ECOLOGICAL ANALYSES OF RNA VIRAL DATA SETS

Concepts and definitions

1. Abundance
2. Alpha diversity
3. Beta diversity

Viral read abundance

$$\frac{\text{Number of reads aligning to RNA viruses}}{\text{Total number of reads in the library}} \times 100$$

Relative viral abundance

$$\frac{\text{Number of reads aligning to a viral taxa}}{\text{Total number of } \textbf{viral} \text{ reads in the library}} \times 100$$

Alpha diversity | diversity *within* a community

- 'Species' richness
- Shannon diversity
- Effective Shannon

- Integer number of unique taxa in a community
- Not restricted to 'species'
- Group by family or higher for diverse viromes

ECOLOGICAL ANALYSES OF RNA VIRAL DATA SETS

Concepts and definitions

1. Abundance
2. Alpha diversity
3. Beta diversity

Viral read abundance

$$\frac{\text{Number of reads aligning to RNA viruses}}{\text{Total number of reads in the library}} \times 100$$

Relative viral abundance

$$\frac{\text{Number of reads aligning to a viral taxa}}{\text{Total number of } \textbf{viral} \text{ reads in the library}} \times 100$$

Alpha diversity | diversity *within* a community

- 'Species' richness
- Shannon diversity
- Effective Shannon

- Add evenness to richness
- Many species, evenly distributed = high H'
- Many species, subset dominates = low H'

ECOLOGICAL ANALYSES OF RNA VIRAL DATA SETS

Concepts and definitions

1. Abundance
2. Alpha diversity
3. Beta diversity

Viral read abundance

$$\frac{\text{Number of reads aligning to RNA viruses}}{\text{Total number of reads in the library}} \times 100$$

Relative viral abundance

$$\frac{\text{Number of reads aligning to a viral taxa}}{\text{Total number of } \textbf{viral} \text{ reads in the library}} \times 100$$

Alpha diversity | diversity *within* a community

- ‘Species’ richness
- Shannon diversity
- Effective Shannon

- Exponent of Shannon index
- Number of equally common species to result in Shannon index of that community

ECOLOGICAL ANALYSES OF RNA VIRAL DATA SETS

Concepts and definitions

1. Abundance
2. Alpha diversity
3. Beta diversity

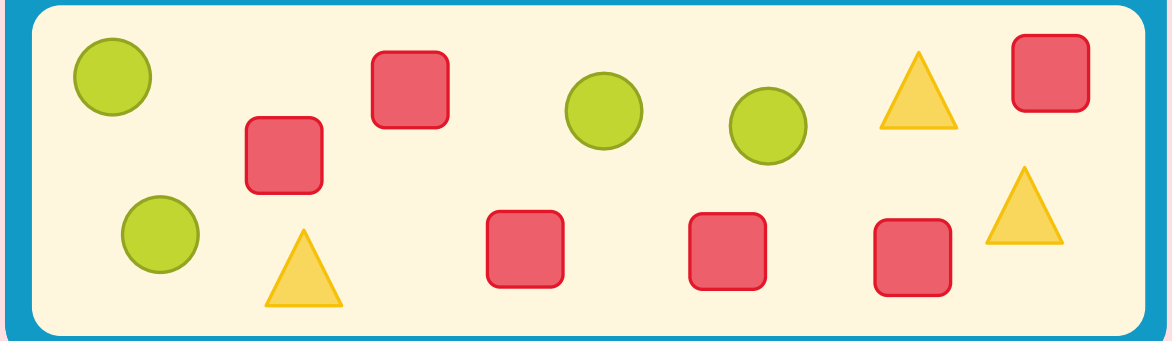
Viral read abundance

$$\frac{\text{Number of reads aligning to RNA viruses}}{\text{Total number of reads in the library}} \times 100$$

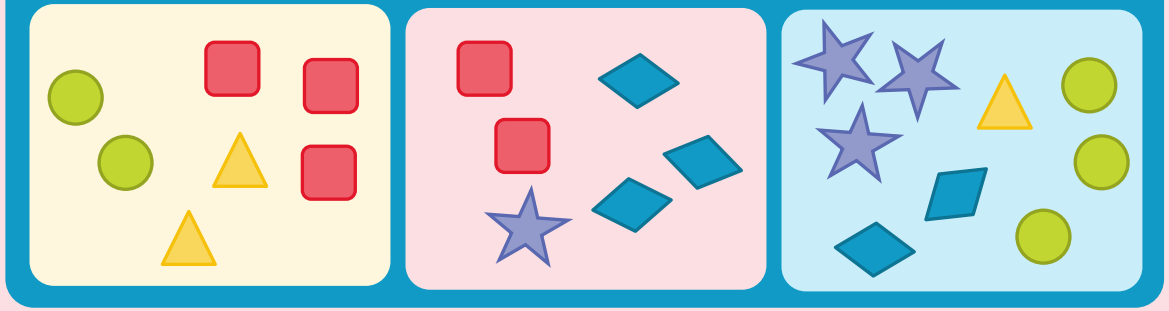
Relative viral abundance

$$\frac{\text{Number of reads aligning to a viral taxa}}{\text{Total number of } \textbf{viral} \text{ reads in the library}} \times 100$$

Alpha diversity | diversity *within* a community



Beta diversity | diversity *between* communities



ECOLOGICAL ANALYSES OF RNA VIRAL DATA SETS

Example data sets

Example data set 1 | *36 libraries*

Soil collected from Liverpool Plains farmlands

3 soil types	Chromosol	Vertosol	Sodosol
2 land uses	Native	Agricultural	
2 depths from surface	0-5 cm	5-15 cm	

Example data set 2 | *32 libraries*

Soil collected from Camden campus in duplicate

2 collection dates	April	September		
Temperatures	-80°C	2-8°C	-30°C	-80°C
Time points	2, 4, 8, 12 weeks	1, 5, 10, 15 days		

Quick glimpse of your data set

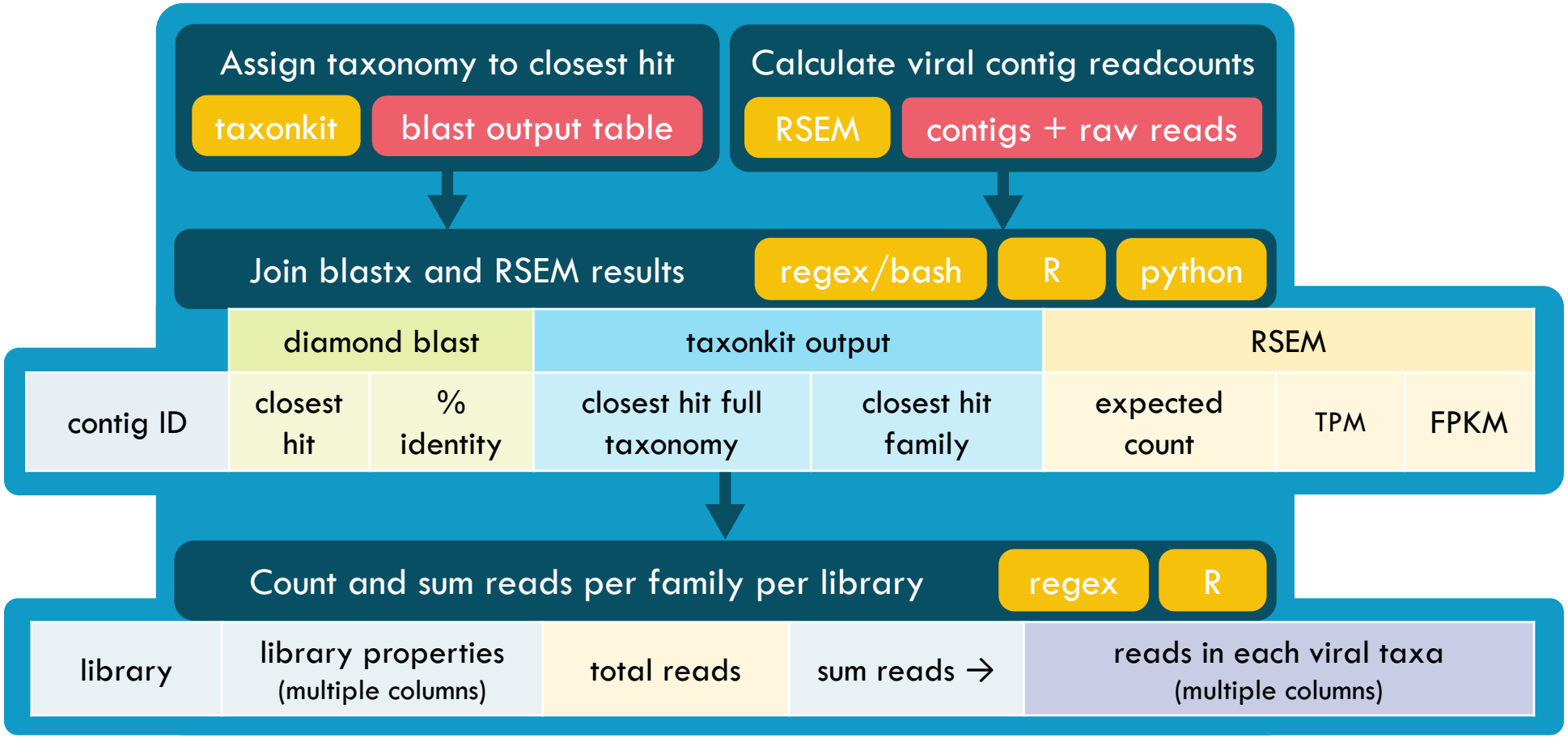
- Virome composition (1 figure)
- Abundance & diversity (1 figure)

Detailed ecological statistics

- Ecological properties vs abundance and alpha diversity
- Best fit model finding
- Beta diversity analysis

METADATA TABLE READCOUNTS/FAMILY/LIBRARY

Setting up your metadata



METADATA TABLE READCOUNTS/FAMILY/LIBRARY

Viral read abundance

$$\frac{\text{Number of reads aligning to RNA viruses}}{\text{Total number of reads in the library}} \times 100$$

Relative viral abundance

$$\frac{\text{Number of reads aligning to a viral taxa}}{\text{Total number of } \textbf{viral} \text{ reads in the library}} \times 100$$

library	library properties (multiple columns)	total reads	sum reads →	reads in each viral taxa (multiple columns)

METADATA TABLE READCOUNTS/FAMILY/LIBRARY

Viral read abundance

$$\frac{\text{Number of reads aligning to RNA viruses}}{\text{Total number of reads in the library}} \times 100$$

Relative viral abundance

$$\frac{\text{Number of reads aligning to a viral taxa}}{\text{Total number of } \textbf{viral} \text{ reads in the library}} \times 100$$

library	library properties (multiple columns)	total reads	sum reads →	reads in each viral taxa (multiple columns)

TODAY'S WORKSHOP AGENDA

Concepts and definitions

Example data sets

Setting up your metadata

- Final list of viral contigs
- Blast results (taxonomy) + RSEM (readcounts)
- Library properties

DONE

Quick glimpse of your data set

- Virome composition (1 figure)
- Abundance & diversity (1 figure)

ViromeComp.csv

Detailed ecological statistics

- Ecological properties vs abundance and alpha diversity
- Best fit model finding
- Beta diversity analysis

metadata.csv

NOTES ON OUR EXAMPLE DATA

Setting up your metadata

ViromeComp.csv

metadata.csv

- ViromeComp.csv has some taxa condensed
  - *Picobirnaviridae* + *Partitiviridae* + *Durna-like* = Durnavirales
  - *Dicistroviridae* + *Marnaviridae* + *Picorna-like* = Picornavirales
- ViromeComp.csv has alpha diversity indices as variables
  - Calculated from metadata.csv
  - ViromeComp.csv chronologically analysed after metadata.csv
  - I'll get to it dw :)

VIROME COMPOSITION LINES 38-113

Quick glimpse of your data set

- Virome composition (1 figure)
- Abundance & diversity (1 figure)

```
37  setwd("/path/to/file/Diversity_Resources/example1")
38  virome1 <-  read.csv("ViromeComp1.csv",na.strings=c("", "NA"), header=TRUE,sep=",")
39
40  ##########################################################################################
41
42  rel1<-melt(virome1, id.vars=c("Library", "Environment", "Soil_type", "Land_use_broad", "Land_use", "Depth",
43                                "Richness", "Shannon", "Shannon.effective", "TotalReads", "SumReads"),
44             variable.name="Family", value.name="Count")
45
46  rel1$Count[is.na(rel1$Count)] <- 0
47  rel1$Pct_rel1<-(rel1$Count/rel1$SumReads)*100
```

Script explanation

Imports and formats data

Calculates relative viral abundances for each library

VIROME COMPOSITION LINES 38-113

v1_RelAbundFam.pdf

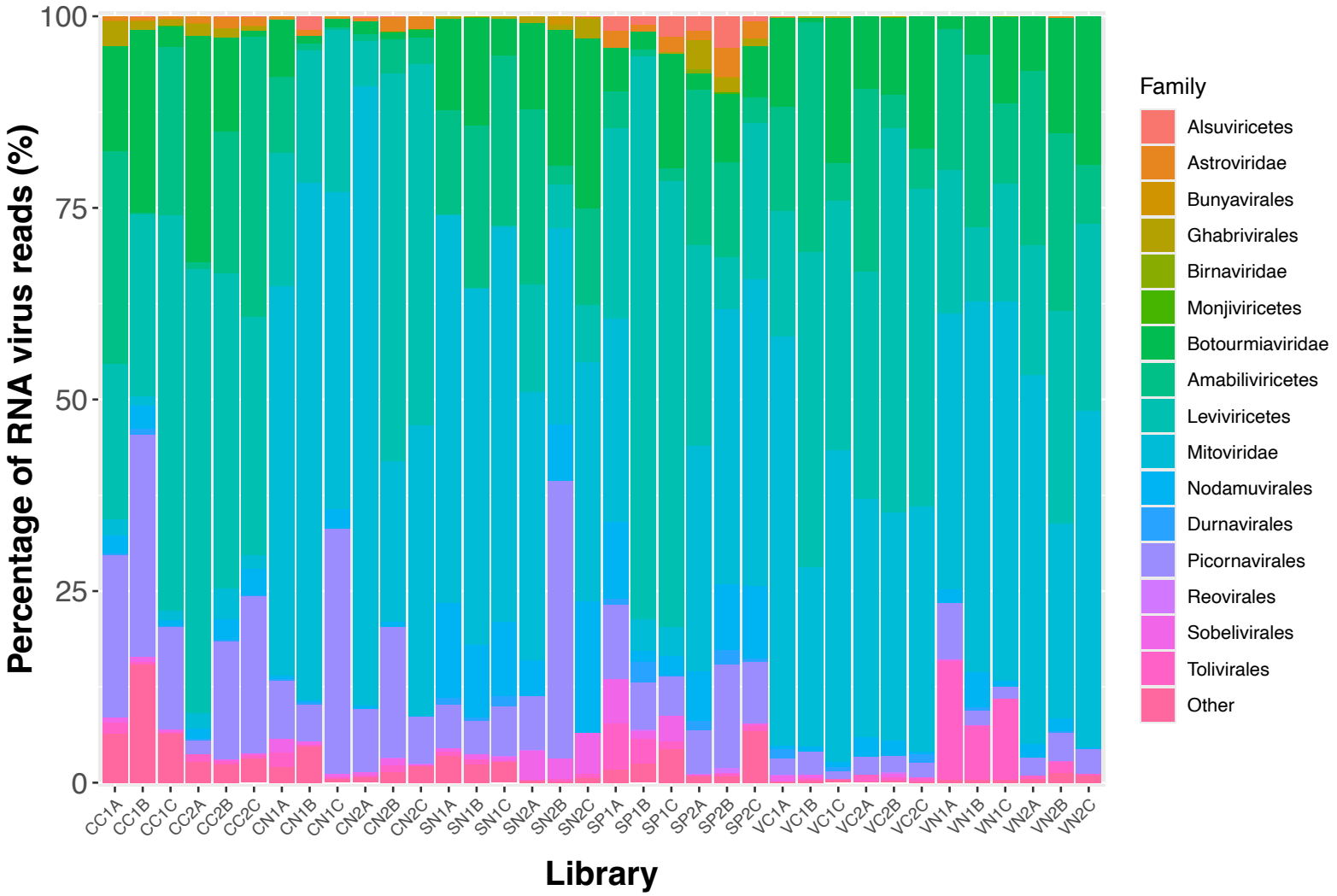
Lines 49-62

- Basic ggplot stuff
- Output →

- Bad colours
- Uninformative x-axis

Lines 65-113

- Improve formatting
- Group by variables



VIROME COMPOSITION LINES 38-113

Script explanation

- Re-order variables (preference)
- "Facets" group libraries by soil type (outer) and land use (inner)

```
65  rel1$Soil_type <- factor(rel1$Soil_type, levels = c("Chromosol", "Vertosol", "Sodosol"))
66  rel1$Land_use <- factor(rel1$Land_use, levels = c("Cropping", "Pasture", "Native"))
67  rel1$Depth <- factor(rel1$Depth, levels = c("5 cm", "15 cm"))
```

~ ggplot magic ~

```
110  z1<-v2 + facet_nested(cols= vars(Soil_type, Land_use), scales = "free_x", space = "free_x") +
111    theme(strip.text.x = element_text(size = 12),
112          legend.position = "right")
113  ggsave("z1_RelAbundFam_facet.pdf", width = 10, height = 7, units = "in")
```

Lines 65-113

- Improve formatting
- Group by variables

VIROME COMPOSITION LINES 38-113

z1_RelAbundFam_facet.pdf

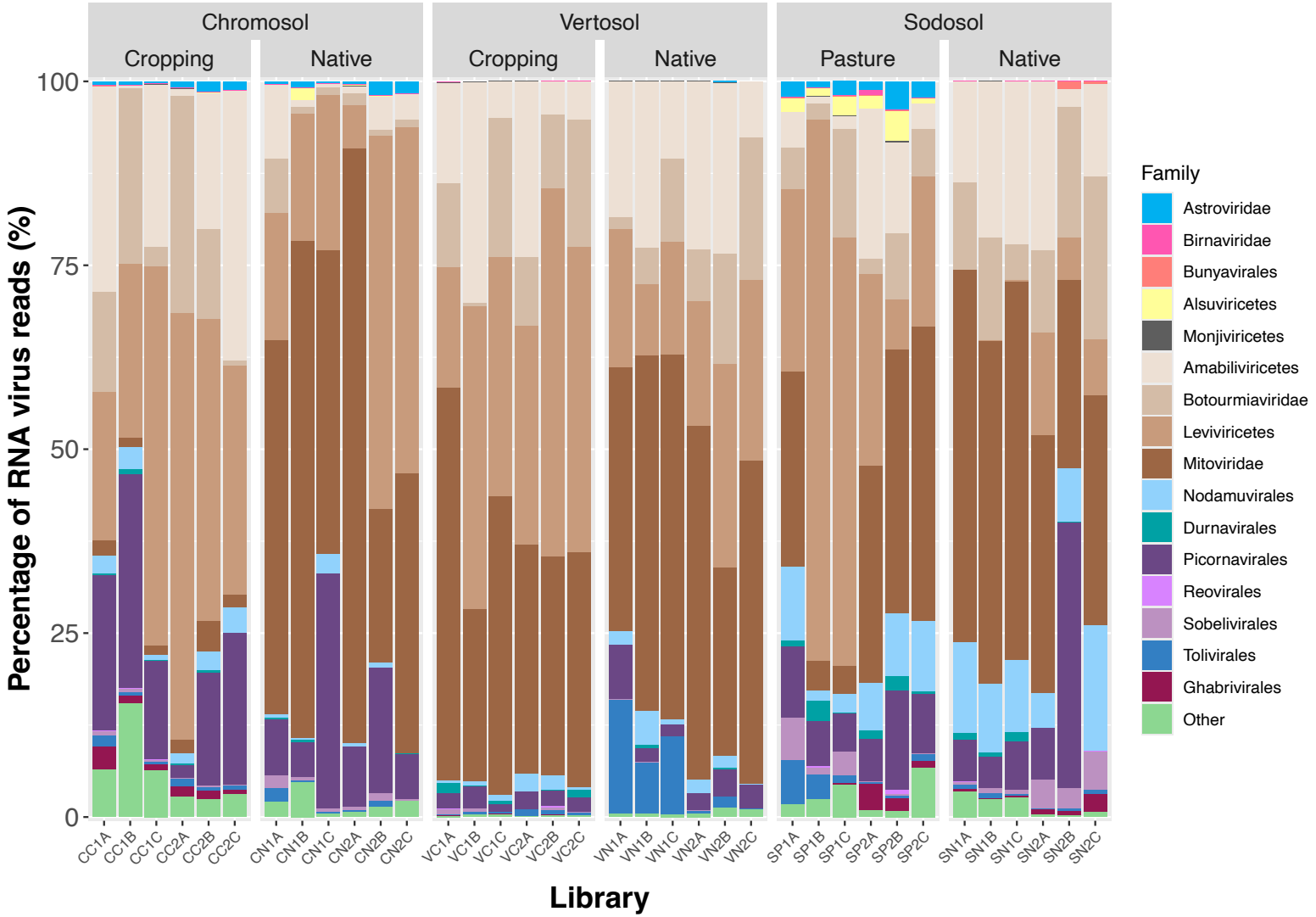
```
65  rel1$Soil_type <- factor(rel1$Soil
66  rel1$Land_use <- factor(rel1$Land_
67  rel1$Depth <- factor(rel1$Depth, l
```

~ ggplot magic ~

```
110  z1<-v2 + facet_nested(cols= vars(Soi
111    theme(strip.text.x = element_text(
112          legend.position = "right")
113  ggsave("z1_RelAbundFam_facet.pdf", w
```

Lines 65-113

- Improve formatting
- Group by variables



ABUNDANCE & DIVERSITY LINES 117-196

Quick glimpse of your data set

- Virome composition (1 figure)
- Abundance & diversity (1 figure)

```
117 dirt1<-melt(virome1, id.vars=c("Library", "Environment", "Soil_type", "Land_use_broad", "Land_use", "Depth",
118                              "Richness", "Shannon", "Shannon.effective", "TotalReads", "SumReads"),
119             variable.name="Family", value.name="Count")
120
121 dirt1$Count[is.na(dirt1$Count)] <- 0
122 dirt1$Abundance<-(dirt1$Count/dirt1$TotalReads)*100
123
124 Abund<-aggregate(Abundance ~ Library + Environment + Soil_type + Land_use_broad + Land_use + Depth +
125                    Richness + Shannon + Shannon.effective + TotalReads + SumReads, data=dirt1, FUN=sum)
126
127 # Re-ordering, same as before
128 Abund$Soil_type <- factor(Abund$Soil_type, levels = c("Chromosol", "Vertosol", "Sodosol"))
129 Abund$Land_use <- factor(Abund$Land_use, levels = c("Cropping", "Pasture", "Native"))
130 Abund$Depth <- factor(Abund$Depth, levels = c("5 cm", "15 cm"))
```

Script explanation

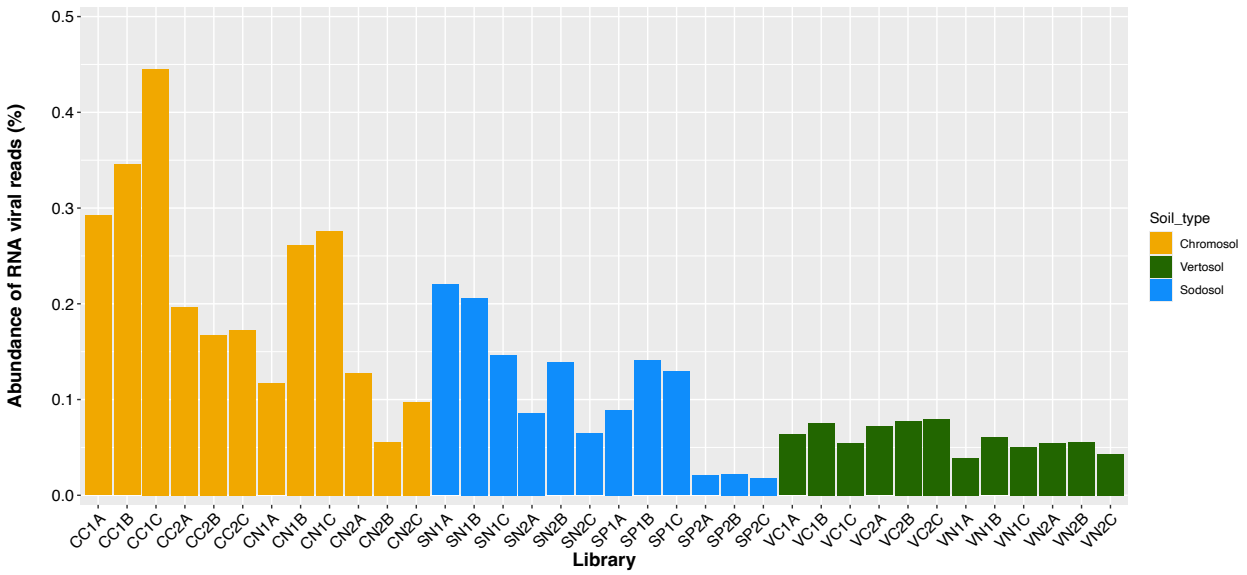
Imports and formats data

Calculates viral read abundance as a proportion of all reads

ABUNDANCE & DIVERSITY LINES 117-196

Lines 132-153

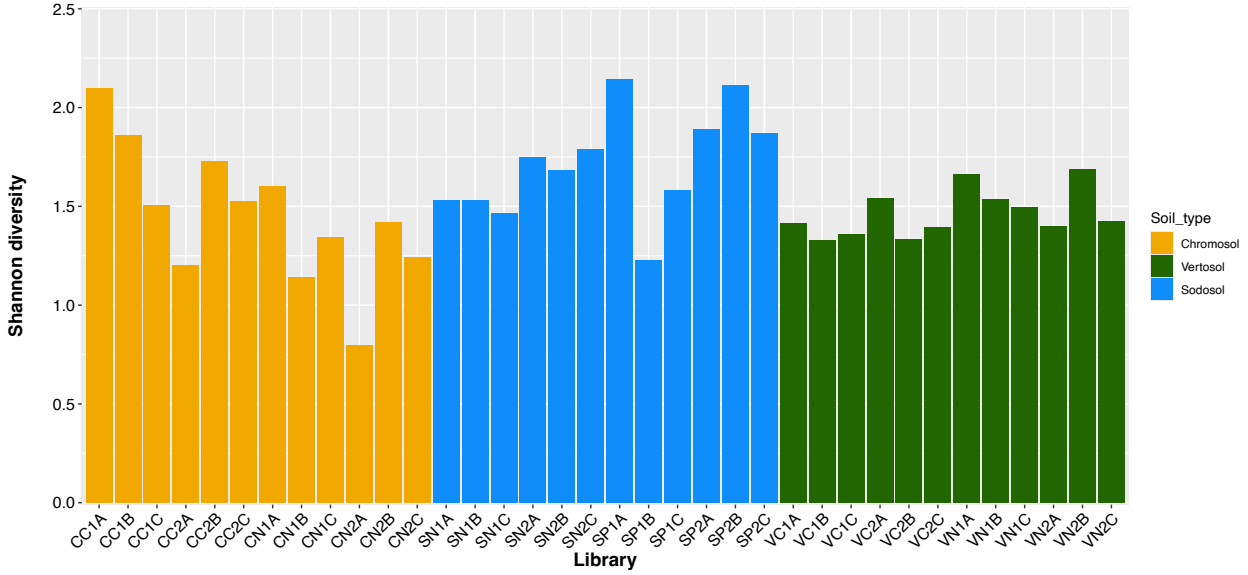
- Plot viral read abundance of each library
- Coloured by soil type



v3_TotalAbund.pdf

Lines 155-176

- Plot Shannon diversity indices of each library
- Coloured by soil type



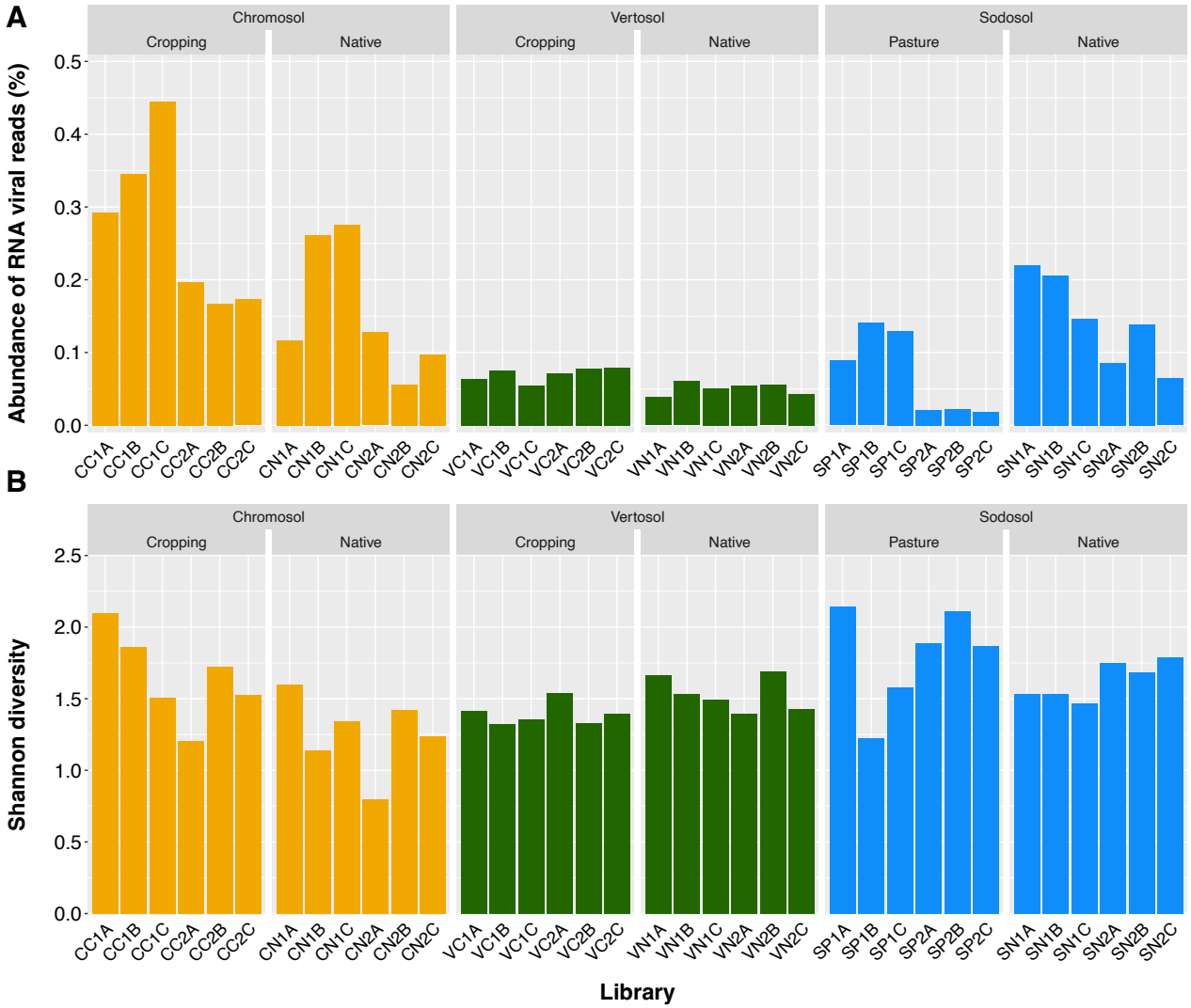
v4_TotalShannon.pdf

ABUNDANCE & DIVERSITY LINES 117-196

v3_TotalAbund.pdf

Lines 178-196

- ggarrange to combine abundance and Shannon diversity figures
- Facet by soil type and land use



ECOLOGICAL STATISTICS IMPORT DATA

Detailed ecological statistics

- Ecological properties vs abundance and alpha diversity

```r
data1 <-  read.csv("metadata1.csv",na.strings=c("", "NA"), header=TRUE,sep=",")

soil1<-melt(data1, id.vars=c("Library", "Environment", "Soil_type", "Land_use_broad", "Land_use", "Depth",
                             "TotalReads", "SumReads"),
            variable.name="Family", value.name="Count")

soil1$Count[is.na(soil1$Count)] <- 0 # Turns empty cells to zeroes
soil1$Abundance<-(soil1$Count/soil1$TotalReads)*100 # Abundance calculated same as  before
soil1$Abundance2<-(soil1$Abundance)*1000000 # For Rhea script - needs to be above 1

Abund<-aggregate(Abundance ~ Library + Environment + Soil_type + Land_use_broad + Land_use + Depth +
                    TotalReads + SumReads, data=soil1, FUN=sum)
```

Script explanation

- Imports and formats data
- Abundance data set = “Abund”

ECOLOGICAL STATISTICS RHEA SCRIPT

Detailed ecological statistics

- Ecological properties vs abundance and alpha diversity

```r
# Calculate the species richness in a sample
Species.richness <- function(x)
{
  # Count only the OTUs that are present >0.5 normalized counts
  count=sum(x[x>1]^0)
  return(count)
}

# Calculate the Shannon diversity index
Shannon.entropy <- function(x)
{
  total=sum(x)
  se=-sum(x[x>0]/total*log(x[x>0]/total))
  return(se)
}

# Calculate the effective number of species for Shannon
Shannon.effective <- function(x)
{
  total=sum(x)
  se=round(exp(-sum(x[x>0]/total*log(x[x>0]/total))),digits =2)
  return(se)
}
```

Our lord and saviour the

✨Rhea script✨

for alpha diversity

(Lagkouvardos *et al*, 2017)

ECOLOGICAL STATISTICS RHEA SCRIPT

Detailed ecological statistics

- Ecological properties vs abundance and alpha diversity

```
250 # DCAST - Remake otu table wide form with Abundance2
251 # CALLS DATA HERE - CHANGE FOR NEW DATA SET ("soil1")
252 otu_table<- dcast(soil1, Family~Library, fun.aggregate = mean, value.var = "Abundance2")
253 otu_table$Family<-NULL
254
255 # Clean table from empty lines
256 otu_table <- otu_table[!apply(is.na(otu_table) | otu_table=="",1,all),]
257 otu_table[is.na(otu_table)] <- 0
258
259 # Order and transpose OTU-table
260 my_otu_table <- otu_table[,order(names(otu_table))]
261 my_otu_table <-data.frame(t(my_otu_table))
262
263 # Apply diversity functions to table#
264 otus_div_stats<-data.frame(my_otu_table[,0])
265 otus_div_stats$Richness<-apply(my_otu_table,1,Species.richness)
266 otus_div_stats$Shannon<-apply(my_otu_table,1,Shannon.entropy)
267 otus_div_stats$Shannon.effective<-apply(my_otu_table,1,Shannon.effective)
268
269 otus_div_stats
270 write.table(otus_div_stats, file="AlphaDiversity1.txt")
271
272 # Add in MetaData
273 # CALLS DATA HERE - CHANGE FOR NEW DATA SET
274 df<-tibble::rownames_to_column(otus_div_stats, "Library")
275 df2<-merge(x = data1, y = df, by = "Library", all = TRUE)
276
277 dflong<-melt(df2, id.vars=c("Library", "Environment", "Soil_type", "Land_use_broad", "Land_use", "Depth", "TotalReads", "SumReads"))
```

Lines 250-277

- Reformat alpha diversity indices into usable structure
- Data set = “df2”

ECOLOGICAL STATISTICS PLOTS TO CHECK

Detailed ecological statistics

- Ecological properties vs abundance and alpha diversity

Lines 281-740

Dependent variables

- Abundance
- Richness
- Shannon
- Effective Shannon

vs

Independent variables

- Environment
- Soil type
- Land use
- Depth

ECOLOGICAL STATISTICS

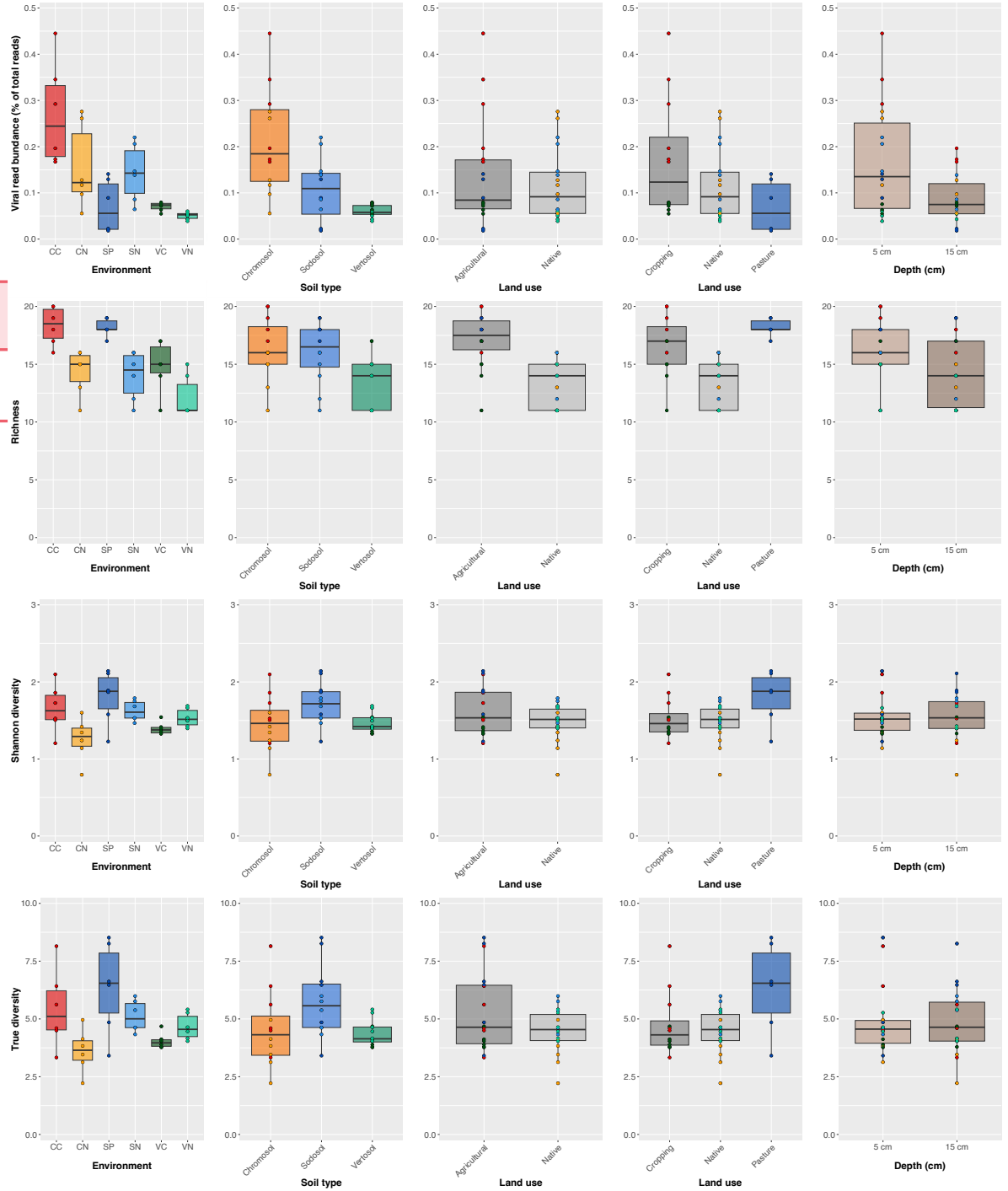
Detailed ecological statistics

- Ecological properties vs abundance and alpha diversity

Lines 281-740

Dependent variables		Independent variables
Abundance	vs	Environment
Richness		Soil type
Shannon		Land use
Effective Shannon		Depth

z3_AllDivByVariables.pdf



ECOLOGICAL STATISTICS

Detailed ecological statistics

- Best fit model finding

Abundance vs each variable individually — anova

If p<0.05, pairwise Tukey tests

Abundance vs combinations of variables — subset regression

Case-specific

Compare model fits to determine best combination — AIC

Lowest AIC wins

Model vs nested model (if needed) — anova

More appropriate for nested *factors*
(but I don't have a published example atm…)

```r
m.env<-lm(Abundance ~ Environment, data=Abund)
print(summary(m.env))
print(anova(m.env,test="Chisq"))

summary(glht(m.env,linfct = mcp(Environment="Tukey")))
```

```r
mmodel_1<-lm(Abundance ~ Environment+Depth, data=Abund)
Anova(mmodel_1, type="II")

mmodel_2<-lm(Abundance ~ Soil_type+Land_use+Depth, data=Abund)
Anova(mmodel_2, type="II")
```

```r
AIC(mmodel_1, mmodel_2, mmodel_3, mmodel_4, mmodel_5,
    m.env, m.type, m.broad, m.use, m.depth)
```

```r
anova(mmodel_1, m.env, test="Chisq")

anova(mmodel_1, m.depth, test="Chisq")
```

ECOLOGICAL STATISTICS

Detailed ecological statistics

- Best fit model finding

Abundance vs each variable individually — anova

If $p<0.05$, pairwise Tukey tests

Abundance vs combinations of variables — subset regression

Case-specific

Compare model fits to determine best combination — AIC

Lowest AIC wins

Model vs nested model (if needed) — anova

More appropriate for nested *factors*
(but I don't have a published example atm…)

Many methods for best subset regression

- Manually compare all models
- Manually drop insignificant variables
- 'dropterm' sequential refining
- 'bestglm' automated model selection

ECOLOGICAL STATISTICS

Detailed ecological statistics

- Best fit model finding

Abundance vs each variable individually — anova

If $p<0.05$, pairwise Tukey tests

Abundance vs combinations of variables — subset regression

Case-specific

Compare model fits to determine best combination — AIC

Lowest AIC wins

Model vs nested model (if needed) — anova

More appropriate for nested *factors*
(but I don't have a published example atm…)

Many methods for best subset regression

- Manually compare all models — Today
- Manually drop insignificant variables
- 'dropterm' sequential refining
- 'bestglm' automated model selection

On Notion

LIVE CODING IN R

Detailed ecological statistics diversity-stats1.R | Lines 744-end

- Ecological properties vs abundance and alpha diversity
- Best fit model finding
- Beta diversity analysis

ECOLOGICAL STATISTICS

Detailed ecological statistics

- Best fit model finding

Dependent variables

- Abundance
- Richness
- Shannon
- Effective Shannon

Significant variables

- Environment
- Soil type
- Land use
- Land use (broad)
- Depth

Best fit model

- Abundance ~ Environment + Depth
- Richness ~ ? + ? + ?
- Shannon ~ ? + ? + ?
- Effective Shannon ~ ? + ? + ?

ECOLOGICAL STATISTICS

Detailed ecological statistics

- Best fit model finding

Dependent variables	Significant variables			Best fit model
Abundance	Environment	Soil type	Depth	Abundance ~ Environment + Depth
Richness	Environment	Soil type	Land use (x2)	Richness ~ Soil type + Land use (broad) + Depth
Shannon	Environment	Soil type	Land use	Shannon ~ Environment
Effective Shannon	Environment	Soil type	Land use	Effective Shannon ~ Environment

HOME STRETCH, TEAM!

Concepts and definitions

Example data sets

Setting up your metadata

- Final list of viral contigs
- Blast results (taxonomy) + RSEM (readcounts)
- Library properties

DONE

Quick glimpse of your data set

- Virome composition (1 figure)
- Abundance & diversity (1 figure)

DONE

Detailed ecological statistics

- Ecological properties vs abundance and alpha diversity
- Best fit model finding
- Beta diversity analysis

BETA DIVERSITY EXAMPLE 2

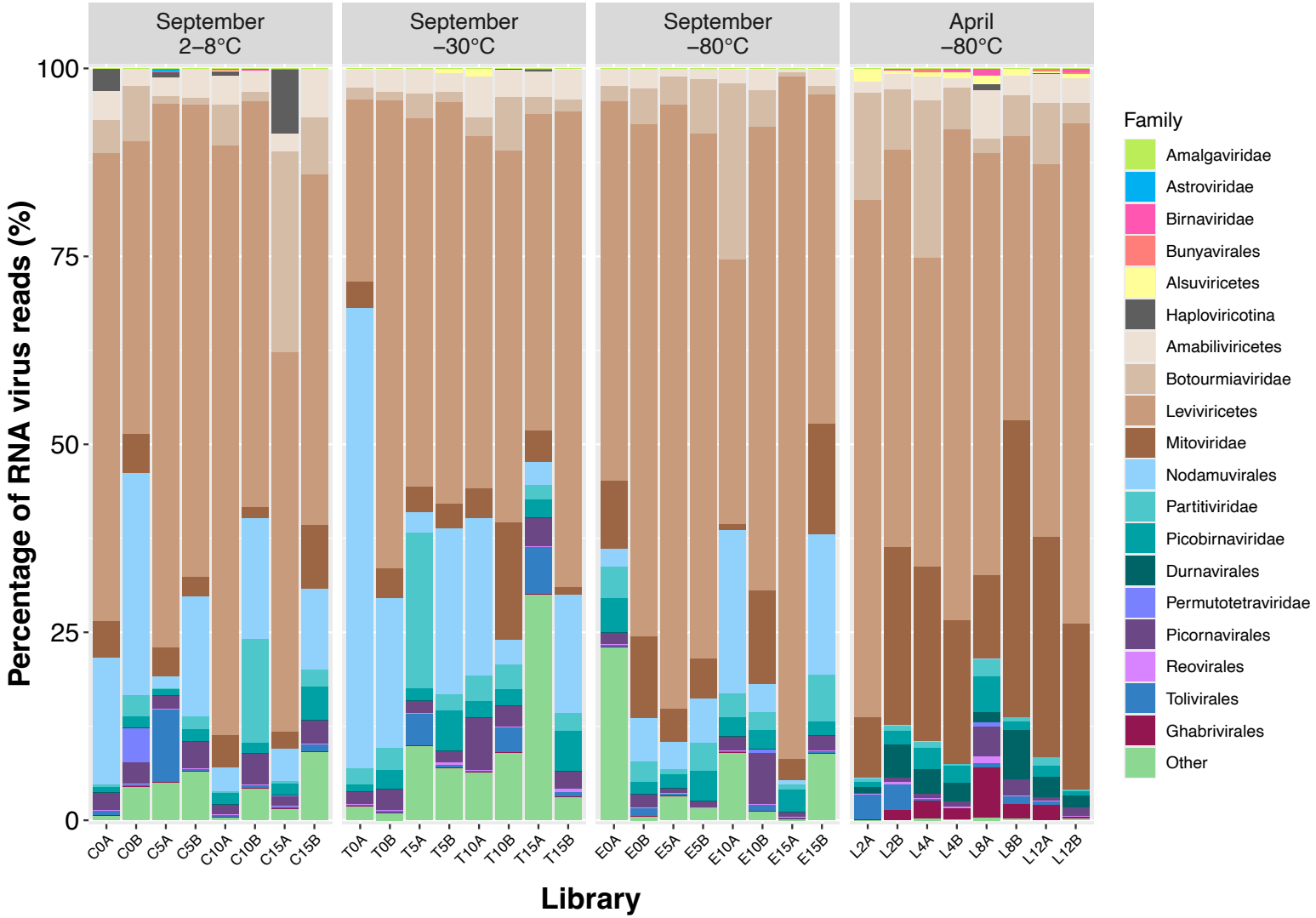
Example data set 2 | *32 libraries*

Soil collected from Camden campus in duplicate

2 collection dates	April	September		
Temperatures	-80°C	2-8°C	-30°C	-80°C
Time points	2, 4, 8, 12 weeks	1, 5, 10, 15 days		

z1_RelAbundFam_facet.pdf

ECOLOGICAL STATISTICS EXAMPLE 2

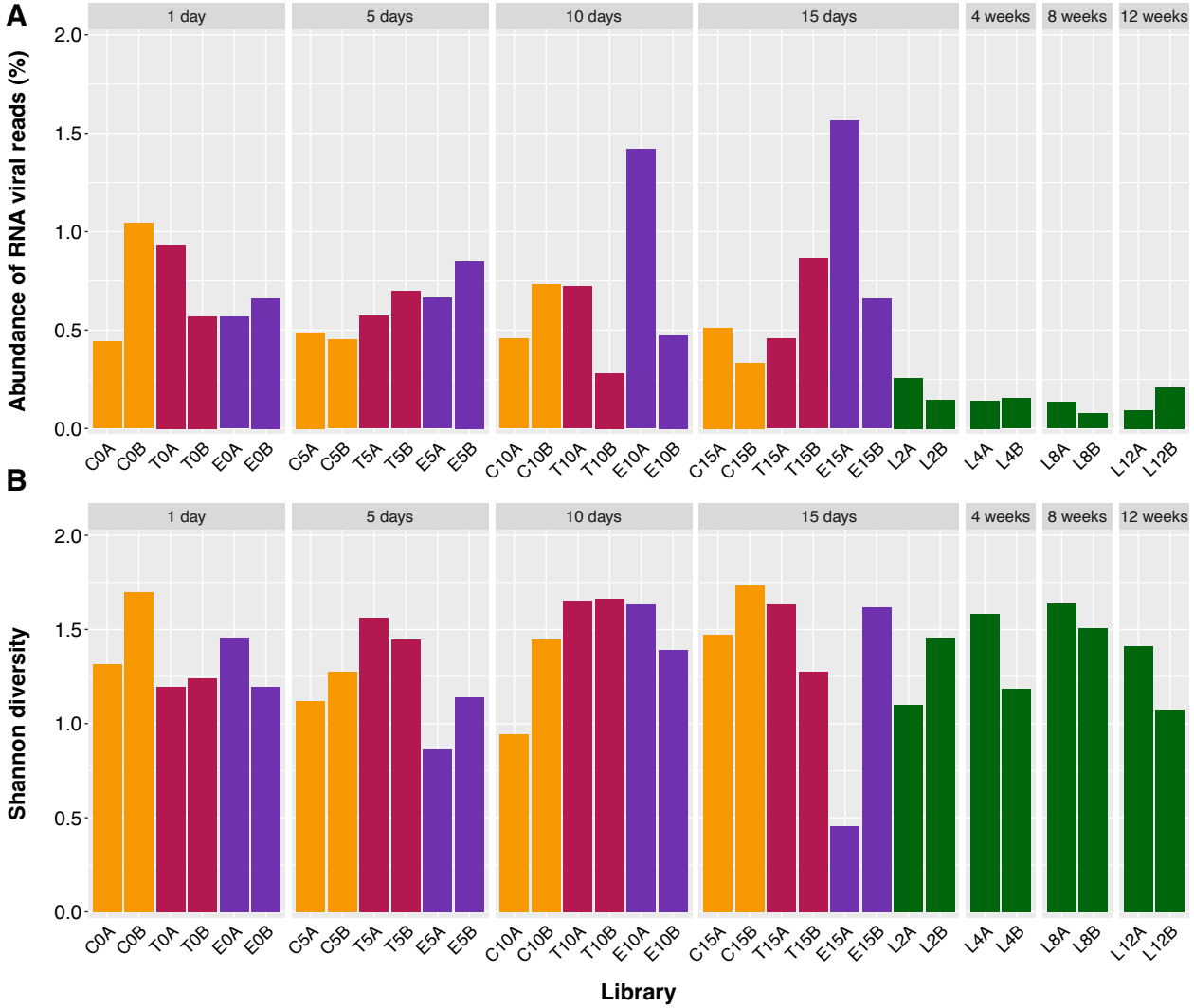


Lines 38-91

- Virome composition

z2_CombinedPlots.pdf

ECOLOGICAL STATISTICS EXAMPLE 2



Lines 38-91

- Virome composition

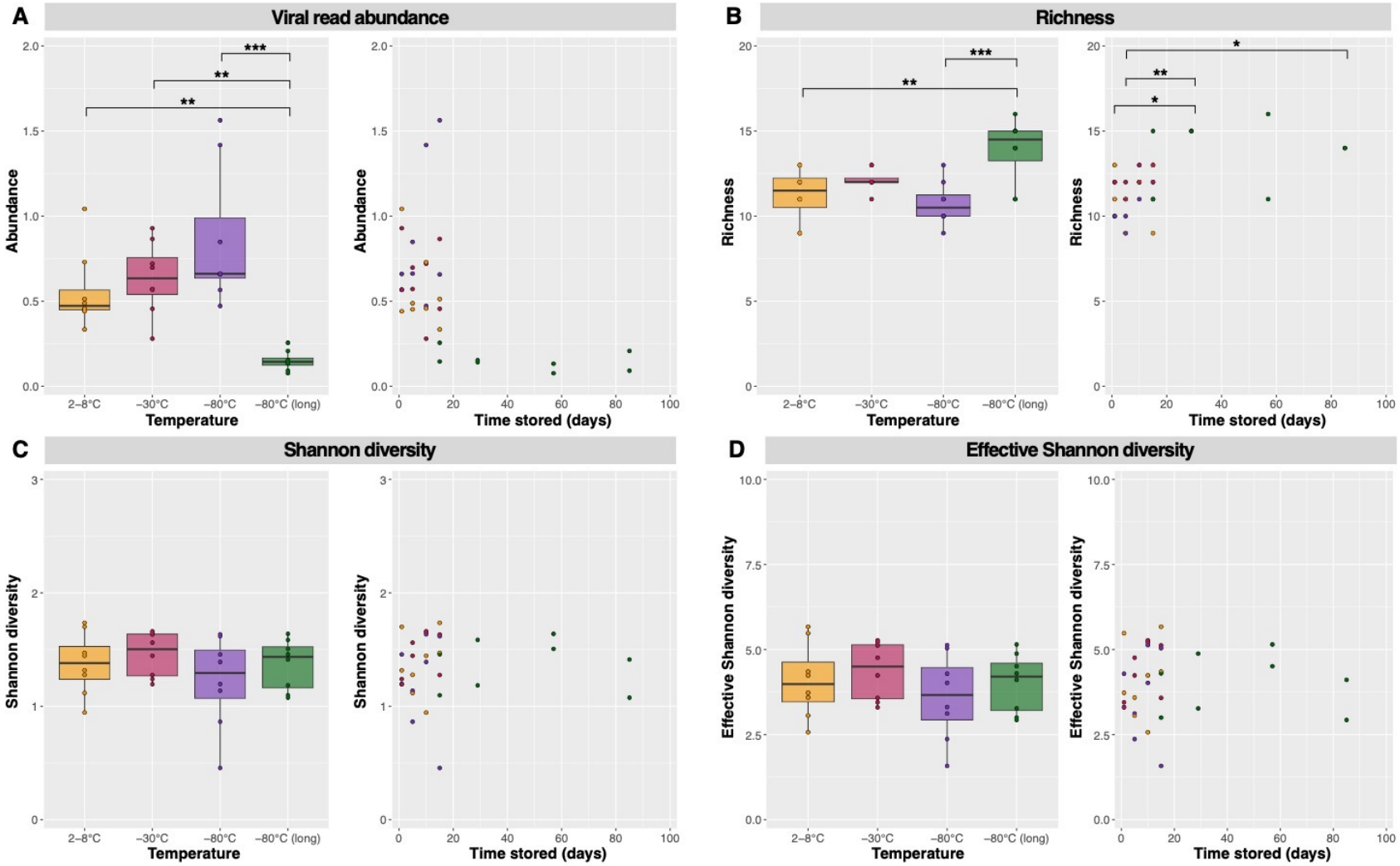
Lines 95-179

- Viral read abundance & Shannon diversity index
- Coloured by storage temp

  (2-8°C -30°C -80°C -80°C)
- Grouped by time stored prior to RNA extraction

z3_DiversityByVariables.pdf

ECOLOGICAL STATISTICS EXAMPLE 2



Lines 38-91

- Virome composition

Lines 95-179

- Viral read abundance & Shannon diversity index
- Coloured by storage temp

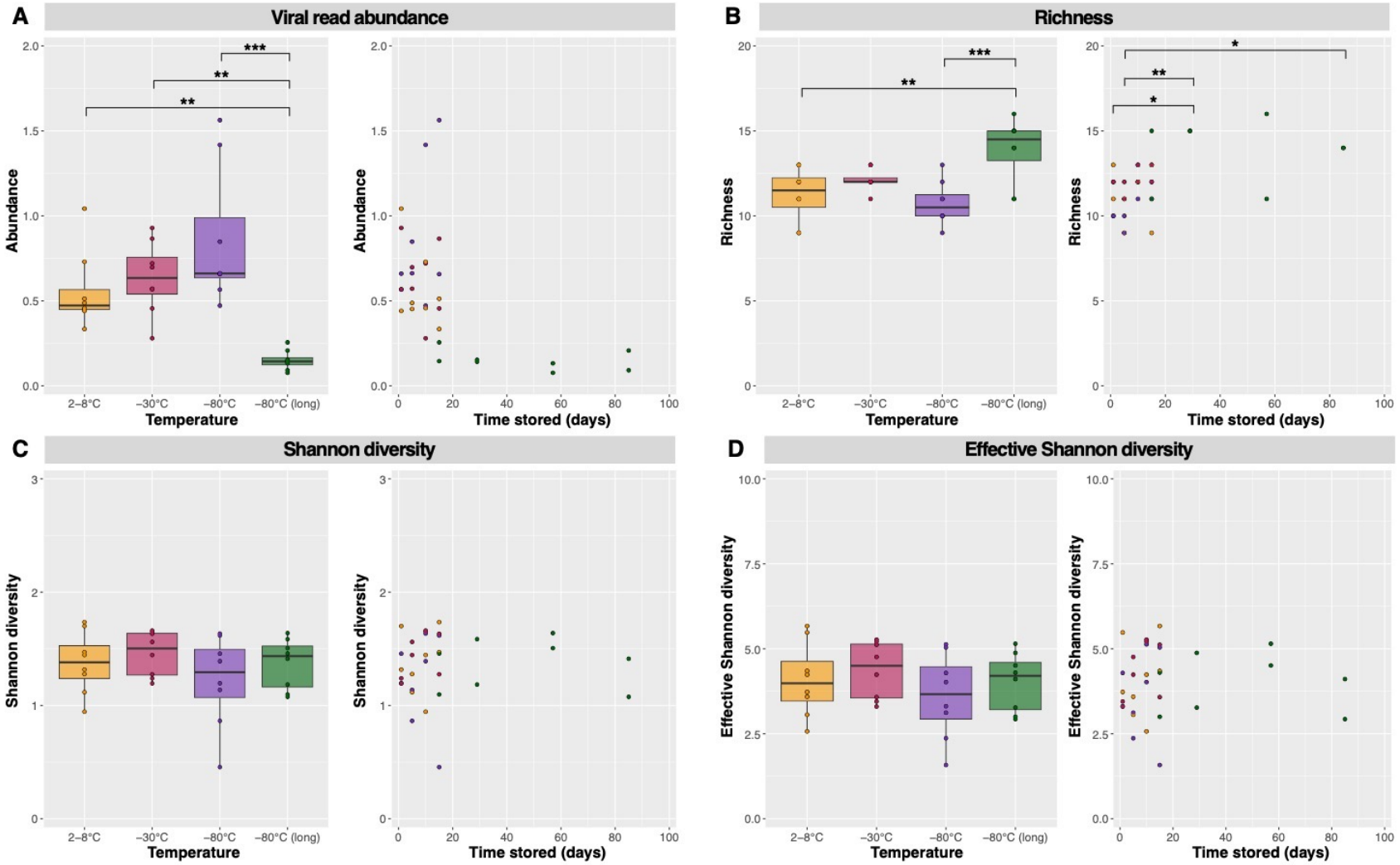
  (2-8°C -30°C -80°C -80°C)
- Grouped by time stored prior to RNA extraction

Lines 183-508

- Abund/alpha div vs variables
- (Cheekily edited in Illustrator)

z3_DiversityByVariables.pdf

ECOLOGICAL STATISTICS EXAMPLE 2



Key findings

- Only pairwise differences are between April and September
- ie, no difference within September samples
- Need to see if variables affect beta diversity…

Lines 183-508

- Abund/alpha div vs variables
- (Cheekily edited in Illustrator)

ECOLOGICAL STATISTICS

Detailed ecological statistics

- Beta diversity analysis

```
516 tmp<- dcast(soil2, Family~Library, value.var = "Abundance")
517
518 # Make a string of the virus families
519 tax<-tmp$Family
520 as.data.frame(tax)
521 TAX<-as.matrix(tax)
522 TAX2 = tax_table(TAX)
523
524 # Make OTU table
525 tmp$Family<-NULL
526 colnames(tmp)<-c("sa1", "sa2", "sa3", "sa4", "sa5", "sa6", "sa7", "sa8",
527                  "sa9", "sa10", "sa11", "sa12", "sa13", "sa14", "sa15", "sa16",
528                  "sa17", "sa18", "sa19", "sa20", "sa21", "sa22", "sa23", "sa24",
529                  "sa25", "sa26", "sa27", "sa28", "sa29", "sa30", "sa31", "sa32")
530 OTU = otu_table(tmp, taxa_are_rows = TRUE)
531
532 # Create a metadata table called "map"
533 map_tmp<-(df2[,c(1:9)])
534 map <- sample_data(map_tmp)
535 str(map)
536
537 # Make it into a phyloseq object
538 physeq = phyloseq(OTU, TAX2, map)
539 str(physeq)
```

Lines 516-539

- Reformats our data into beta diversity-compatible structure
- Just trust me bro

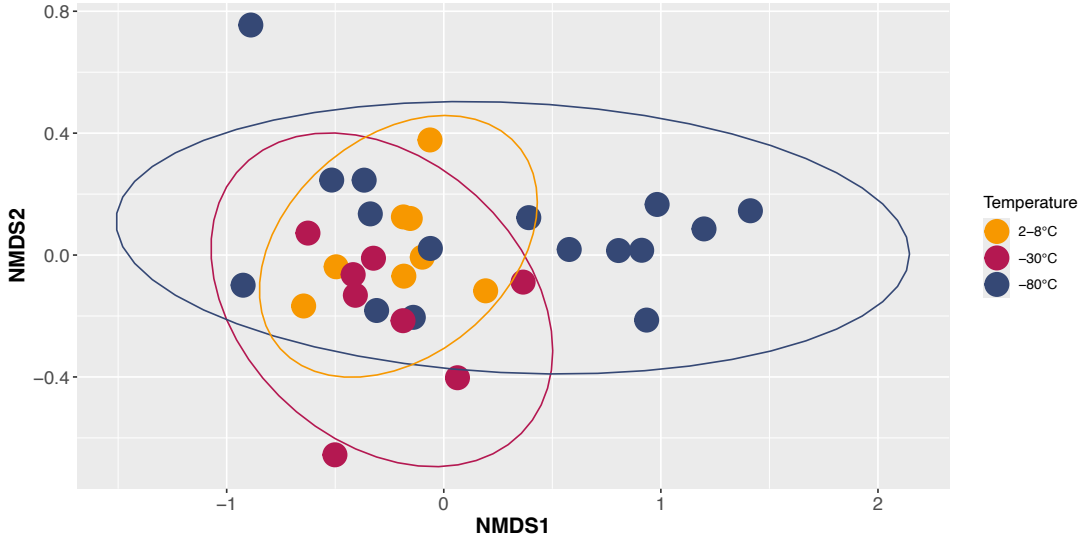
b1_BetaDiversity_nMDS_Temp.pdf

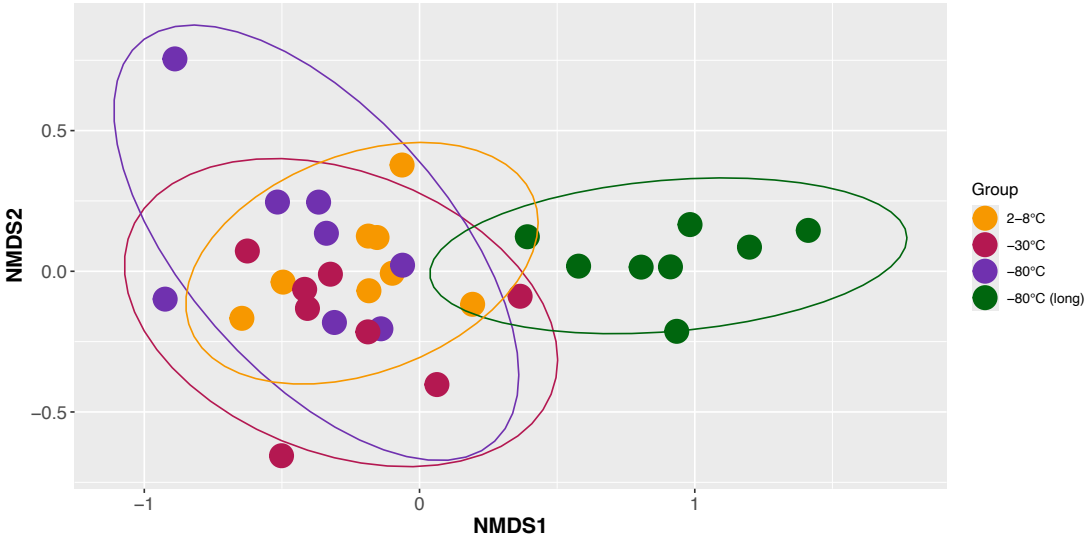
b2_BetaDiversity_nMDS_Group.pdf

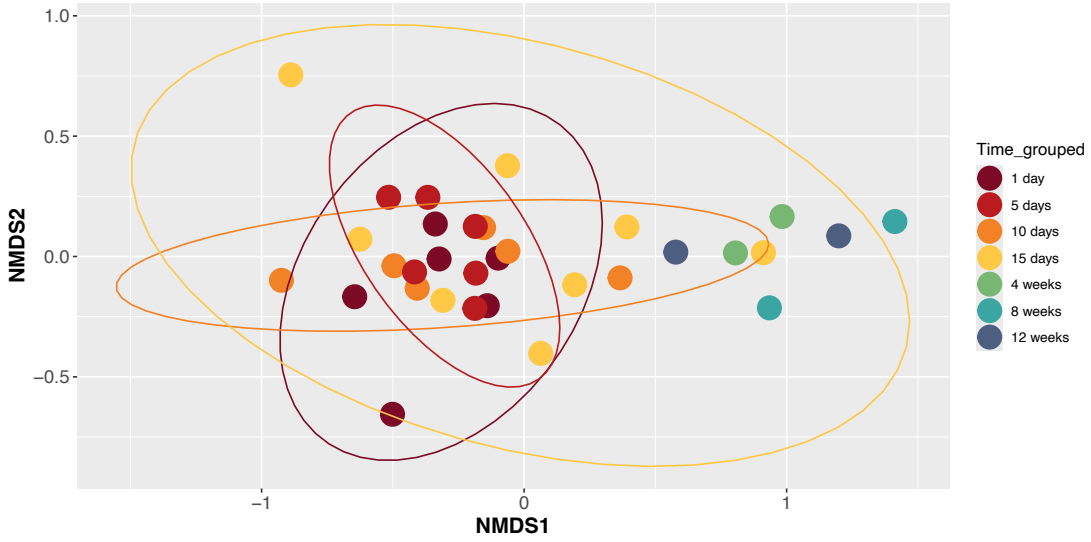
b3_BetaDiversity_nMDS_Time.pdf

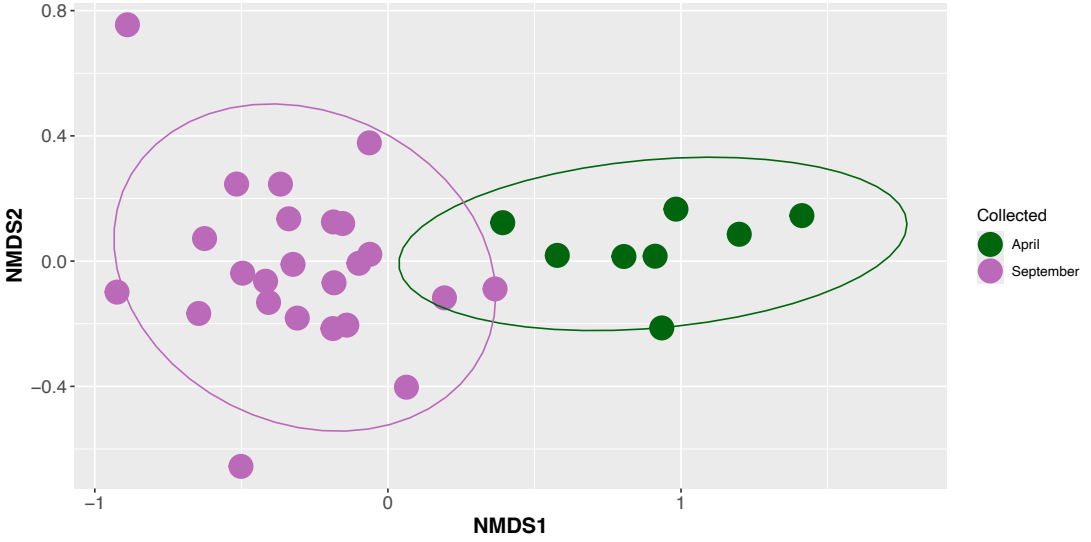
z4_BetaDiversity_nMDS_Month.pdf

BETA DIVERSITY









LIVE CODING IN R

Detailed ecological statistics diversity-stats2.R | Lines 585-end

- Ecological properties vs abundance and alpha diversity
- Best fit model finding
- Beta diversity analysis

HOMEWORK NOT COMPULSORY, JUST SUGGESTIONS

Recommendations for follow up/further practice

- Try the model testing pipeline on example 2 data set
  - Spoiler alert: time and temperature will be very straightforward
  - Several other variables in metadata2.csv (not explored today)
    - You can test them out for practice
- Try adapting example2.R to data set of only *September* samples
  - ie, remove 'L' libraries from metadata2.csv
  - Is there a relationship between storage conditions and virome within samples taken from one collection point?
  - What do you expect to see here?

Example 1 paper
(Sadiq *et al*, 2024)

Example 1 paper
(Sadiq *et al*, 2024)

KEY TAKEAWAY

Not all-encompassing, but a good start

- p-value isn't everything - always go back to biological context

Design projects with ecology in mind | can brute force it but… risky

Make plots of everything | easier to identify trends visually

Interpret models carefully | consider confounding or nested variables

Me what time I finished this workshop

I'll be available to delve deeper, answer questions, and chat about applying this to your projects

Hacky hour next lab meeting ~3pm April 1

Example 1 paper (Sadiq *et al*, 2024)

Example 1 paper (Sadiq *et al*, 2024)